

4: How to measure the air resistance by considering the model point as being fixed to a car moving along a linear s-axis.

Proceeding as described in sec. 3, two sets of well defined coefficients a and β are obtained. How can we use these to determine the air resistance against the model? Assume first that the model had travelled not on merry-go-round, but on a little car as illustrated in fig. 4.1.

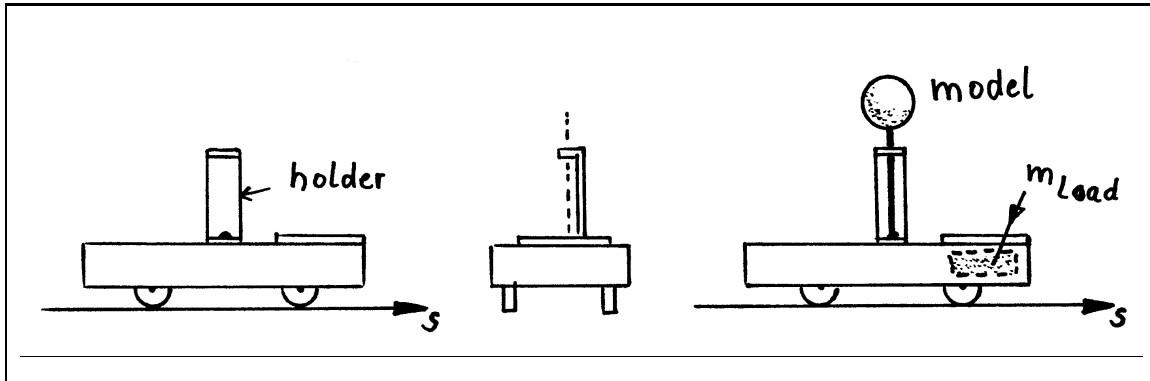


Figure 4.1: A car with mass m , bound to move along a horizontal s -axis. Two experiments are compared: One where it rolls along with a model of mass m_{model} mounted on a holder and one where it rolls without this model. In both cases it loses its speed because of air resistance and a tiny friction in the bearings. To movements in the air caused by car and holder from reach the model, the holder is very thin seen from the front of the car, and it is relatively tall.

An extra mass m_{load} can be placed in a closed room to help stabilising the car when it carries a model.

Assume further that in both runs, the air resistance and the tiny friction in the bearings give the car an acceleration which can be written as

$$a = -a v - \beta v^2, \quad (4.1)$$

where a and β are constants. According to Newton's 2. law, this amounts to assuming that the force against the movement can be written as

$$F_{\text{air,bearings}} = f_1 v + f_2 v^2 \quad \text{where} \quad f_1 = m a, \quad f_2 = m \beta, \quad (4.2)$$

m being the mass. Letting m_e denote the mass of the car when it runs *without* the model (e for *empty*), we have

$$m = m_e + m_{\text{model}} + m_{\text{load}} \quad (4.3)$$

when it runs *with* the model. Assuming finally that the tiny friction in the bearings is the same in the two runs, the formula for the air resistance against the model at velocity v is found to be

$$F = \Delta f_1 \cdot v + \Delta f_2 \cdot v^2, \quad (4.4)$$

where Δf_1 and Δf_2 are the constants found by filling in table 4.1.

Parameter	α	β	m	$f_1 = m\alpha$	$f_2 = m\beta$
Unit	$\frac{1}{s}$	$\frac{1}{m}$	g	$\frac{\text{milliN}}{m/s}$	$\frac{\text{milliN}}{(m/s)^2}$
Running with model	0.002110	0.003684	594	1.254	2.188
Running without model	0.002676	0.002363	557	1.491	1.316
$\Delta f = f_{\text{with}} - f_{\text{without}}$				-0.237	0.872

Tabel 4.1: Table for calculation of the constants Δf_1 and Δf_2 in the formula

$$F = \Delta f_1 \cdot v + \Delta f_2 \cdot v^2$$

for the air resistance. The values of α and β used as an example are those obtained from the fits in figs. 3.3 and 3.4. The corresponding values for the mass m are those obtained by the measurements described in fig. 5.2.

The model used in the example is a smooth sphere with diameter $d = 62.5 \text{ mm}$, namely the ball shown in fig. 6.1 and considered again in figs. 6.2 and 6.3.

Now, the model is *not* carried by a car in its two runs, it is carried by the merry-go-round. The car is a *mathematical model* of the merry-go-round, used only to *simulate* the movement of the model point. To do so correctly,

we shall think of it as having the mass m determined by the experiments described in figs. 5.2a and 5.2b (4.5a)

in the next section. It can then be shown (see appendix B) that

if the merry-go-round moves as the result of one single force F , acting in the model point and directed along the s -axis, the model point will move exactly as the car, when acted upon by the same single force, directed along the linear s -axis (4.5b)

Because of this property, we shall sometimes refer to the car with this mass m as *the equivalent car*. And we shall sometimes call m *the model point mass*. When that is found, we can write the drag coefficient C_d defined in (2.1) as¹

$$C_d = \frac{2F}{\rho A v^2} = \frac{c_1}{v} + c_2 \text{ where } c_1 = \frac{2\Delta f_1}{\rho A} \text{ and } c_2 = \frac{2\Delta f_2}{\rho A} \quad (4.6)$$

are known constants. Note that c_1 is a velocity while c_2 is dimension less.

¹ When the merry-go-round rotates without a model, the air resistance acts all over its entire body. However, its total effect can be thought of as being due to a single force acting in the model point. When the model is mounted, the total effect of the air resistance increases by the effect of the air resistance against the model, which *only* acts on the model point.

Actually objections can be raised about the last assertion. Maybe the presence of the model distorts the flow of air so much, that the merry-go-round does not feel exactly the same resistance as when rotating alone. *Hence it is an assumption for the following, that the airflow around the model only interferes very little with the flow around the merry-go-round.*